

Sprint 2 Documentation – Project Setup, REST API, DAL

Team: Sebastian and Felix

Repository: <https://github.com/BunnySupreme/SWEN3Project>

Sprint duration: 17.10.2025 – 03.11.2025

Estimated workload: 10 h / person / week

1. Sprint Goal (from assignment)

- Set up Vue.js/Quasar frontend project (“paperless-frontend” service) with dashboard, detail pages for each document, and upload page
- Set up nginx server with reverse proxy for backend endpoints
- Extend docker-compose.yml to include frontend
- Validation (frontend and backend)
- Provide unit tests for DocumentService and DocumentsController
- AutoMapping for DTOs (leftover from Sprint 1)

2. Implemented Functionality

2.1 Frontend Project Setup

- Frontend reachable via <http://localhost> and <http://localhost:80>
- Upload saves file info to DB
- Uploaded files info from DB is display in dashboard
- Details about files are viewable per file in dashboard
- Frontend validation prevents user from uploading files with invalid filetypes and sizes, as well as exceeding character and count limits for title and tags

2.2 Webserver (nginx)

For the setup of the web server, see the separate documentation file under [documentation/Webserver.pdf](#)

2.3 Backend Project

- AutoMapping was added for DTOs to Models and Models to DTOs, each where required
- Backend validation was added for each DTO used for data received from users

3. Architecture (Sprint 2)

- Presentation layer: Vue.js/Quasar app
- API layer: DocumentsController
- Business layer: DocumentService
- Data access layer: DocumentRepository
- Database: PostgreSQL container from docker-compose.yml

Refer to Architecture.md for diagrams.

4. Testing

Unit tests implemented in project Paperless.UnitTests:

- DocumentsControllerTests
- DocumentServiceTests
- DocumentServiceValidationTests

Database, repositories, validators, mappers and services mocked using Moq, each where required

All tests passing at the end of Sprint 2

5. Problems

AutoMapping turned out to need a different approach for Model to DTO mapping than for DTO to Model mapping. Constructor parameters had to be engaged with directly, otherwise the set values would simply be ignored.

6. Workload (Estimate)

Team Member	Week / Sprint	Estimated Hours	Main Tasks
Sebastian	Sprint 1	12	CI/CD pipeline, Dockerfile, docker compose yaml, Validators, AutoMapper, Webserver
Felix	Sprint 1	12	Vue.js/Quasar App, Frontend Validation, Controller, Endpoints, BL, DTO